Qn. Set Code-7(SOT)

Semester: 5th
Programme: B.Tech
Branch: CSE, CE, EE

AUTUMN END SEMESTER EXAMINATION-2024

5th Semester B.Tech

ENGINEERING ECONOMICS

HS30101 / HS_2002

(For 2023 (L.E), 2022 & Previous Admitted Batches)

Time: 2 Hours 30 Minutes | Full Marks: 50

Answer any FIVE questions.
Question paper consists of two SECTIONS i.e. A and B.
Section A is compulsory.
Attempt any Four question from Sections B.
The figures in the margin indicate full marks.
Candidates are required to give their answers in their own words as far as practicable and all parts of a question should be answered at one place only.

SECTION-A

1. Answer the following questions. [1 × 10]

(a) Explain with the help of demand curve how increase in demand is different from contraction in demand.

(b) A consumer purchases 20 units of a commodity when his income is ₹ 50,000 per month. The consumer purchases 15 units of the commodity when his income increases to ₹ 70,000. Find the income elasticity of demand for the commodity and mention the type of the good with specific reason.

(c) Shyam needs ₹ 20,00,000 after 15 years. Find out what equal amount of money he should deposit at the end of every year to get ₹ 20,00,000 after 15 years if the interest rate is 12% compounded annually.

(d) Write an expression to find the effective rate of interest.

(e) Define increasing returns to scale.

(f) Diagrammatically show the relationship between Marginal Cost (MC) and Average Variable Cost (AVC).

(g) What is margin of safety?

(h) Write an expression for calculating the depreciation factor in case of sum-of-year digit method of depreciation.

(i) What is demand pull inflation?

(j) What do you mean by Cash Reserve Ratio (CRR)?

SECTION-B

2. (a) Explain the degrees of price elasticity of demand with the help of demand curves. [5]

(b) From the following demand function find out Total Revenue (TR) function and Marginal Revenue (MR) function. [5]

$Q = 80,000 - 10P$

(i) At what price and quantity will TR be maximum?

(ii) Calculate the value of MR when TR is maximum.

3. (a) A company has the following short-run production function: $Q = 9Y^2 - 0.6Y^3$ (Where Q is the output and Y is the variable input) [5]

Find out:

(i) Marginal product function

(ii) Average product function

(iii) Value of Y at which output will be maximum?

(iv) Value of Y at which average product and marginal product will be maximum?

(b) What is an Iso-Cost line? Explain the producer equilibrium point with the help of isoquant and Isocost lines for output maximization. [5]

4. (a) A company has a fixed cost of ₹ 5000. The Total Variable Cost (TVC) function of the company is TVC $= 3000Q - 25Q^2 + 0.5Q^3$. [5]

Find out:

(i) Average Variable Cost (AVC) function

(ii) Average Cost (AC) function

(iii) Marginal Cost (MC) function

(iv) Level of output at which MC and AVC will be minimum

(b) A company sets the selling price of its product per unit at₹ 50. The variable cost per unit and fixed cost of the company are ₹ 30 and 80,000 respectively. [5]

Find out:

(i) Break Even Point (BEP) in quantity and in sales value

(ii) P/V ratio

(iii) Margin of safety if the profit of the company is ₹1,00,000.

5. (a) A company has to select the best project out of two alternative projects. From the following data find out which project should be selected on the basis of the present worth method if i=12% compounded annually. [5]

Particulars	Project A	Project B
Initial Cost (in ₹)	20,00,000	25,00,000
Annual Equal Return (in ₹)	2,00,000	2,50,000
Salvage Value (in ₹)	30,000	40,000
Life (in years)	25	25

(b) A company has to select the best project out of two projects. Project 1 requires an initial cost of ₹ 40,00,000 with an estimated annual equal return of ₹ 2,00,000 every year for the next 6 years. Project 2 requires an initial cost of ₹ 70,00,000 with an estimated annual equal return of ₹ 5,00,000 every year for the next 6 years. Find out which project the company should select using the BC ratio value if the interest rate is 15% compounded annually(Use present worth method). [5]

6. (a) A machine has been purchased with ₹ 2,00,00 with an estimated salvage value of ₹ 40,000. Find out the depreciation amount and book value of the machine for each year by the sinking fund method of depreciation if the interest rate is 10% compounded annually. [5]

(b) Explain the fiscal policy measures taken by the Government for controlling inflationary pressure in an economy. [5]

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